# Integration Problems

1. $\int \frac{dx}{x^2}$

2. $\int_1^2 \frac{dx}{(2x-1)\sqrt{2x-1}}$

3. $\int_e^{e^2} \frac{\log x}{x} dx$

4. $\int x^2 e^{-x} dx$

5. $\int (\log x)^2 dx$

6. $\int x sin^2 x dx$

7. $\int \frac{sinx - cosx}{sinx + cosx} dx$

8. $\int \sqrt{x} \log \sqrt{x} dx$

# Answers

1. $= -\frac{1}{x} + C$

2. $= 1 - \frac{1}{\sqrt{3}}$

3. $= \frac{3}{2}$

4. $= -(x^2 + 2x + 2)e^{-x} + C$

5. $= x(logx)^2 - 2xlogx + 2x + C$

6. $= \frac{x^2}{4} - \frac{x}{4}sin2x - \frac{1}{8}cos2x + C$

7. $= -log|sinx + cosx| + C$

8. $= \frac{1}{3}x^{\frac{3}{2}}logx - \frac{2}{9}x^{\frac{3}{2}} + C$

# Integration Problems

1. $\int_0^1 \frac{1-x}{\sqrt{x}+1}dx$

2. $\int \frac{dx}{(x-2)(x+1)^2}$

3. $\int_1^e x^3 \log x dx$

4. $\int_e^{e^2} \frac{\log x}{x}dx$

5. $\int \frac{e^x - e^{-x}}{e^x + e^{-x}}dx$

6. $\int sin^3\theta cos^2\theta d\theta$

7. $\int xcos^3 x dx$

8. $\int (sin^4 x + \cos^4 x)dx$

# Answers

1. $= \frac{1}{3}$

2. $= \frac{1}{9} log|\frac{x-2}{x+1}| + \frac{1}{2(x+1)} + C$

3. $= \frac{3e^4+1}{16}$

4. $= \frac{3}{2}$

5. $= log(e^x + e^{-x}) + C$

6. $= -\frac{cos^3\theta}{3} + \frac{cos^5\theta}{5} + C$

7. $= \frac{x}{4}(\frac{1}{3}sin3x + 3sinx) + \frac{1}{4}(\frac{1}{9}cos3x + 3cosx) + C$

8. $= \frac{1}{4}(3x + \frac{1}{4}sin4x) + C$

# Integration Problems

1. $\int \sqrt{x^2+1}dx$

2. $\int (x^2+2x+3)(x+1)dx$

3. $\int x(\log x)^2 dx$

4. $\int_0^1 x\log(x+1)dx$

5. $\int_{-1}^1 (x^2+1)e^{-x}dx$

6. $\int \sin(\pi x)dx$

7. $\int x^2 \sin \pi x dx$

8. $\int \frac{\tan x}{\cos x}dx$

# Answers

1. $= \frac{1}{2}(x\sqrt{x^2+1} + log(x + \sqrt{x^2+1})) + C$

2. $= \frac{1}{4}(x^2 + 2x + 3)^2 + C$

3. $= \frac{x^2}{2}\{(logx)^2 - logx + \frac{1}{2}\} + C$

4. $= \frac{1}{4}$

5. $= 2e - \frac{6}{e}$

6. $= -\frac{1}{\pi}cos\pi x + C$

7. $= (\frac{2}{\pi^3} - \frac{x^2}{\pi})cos\pi x + \frac{2}{\pi^2}xsin\pi x + C$

8. $= \frac{1}{cosx} + C$

# Integration Problems

1. $\displaystyle\int_0^1 \frac{x^5 + 2x^3}{(1+x^3)^3} dx$

2. $\displaystyle\int_0^1 \sqrt{4 - x^2} dx$

3. $\displaystyle\int \frac{1}{x \log x} dx$

4. $\displaystyle\int \frac{1}{1 + e^x} dx$

5. $\displaystyle\int \frac{e^x + 1}{e^x - e^{-x}} dx$

6. $\displaystyle\int sin^3 3\theta cos 2\theta d\theta$

7. $\displaystyle\int \sqrt{x} \log \sqrt{x} dx$

8. $\displaystyle\int_0^{\frac{\pi}{2}} \cos^4 x dx$

# Answers

1. $= \frac{7}{24}$

2. $= \frac{\pi}{3} + \frac{\sqrt{3}}{2}$

3. $= log|logx| + C$

4. $= x - log(e^x + 1) + C$

5. $= log|e^x - 1| + C$

6. $= \frac{1}{4}sin2\theta - \frac{1}{32}sin8\theta - \frac{1}{16}sin4\theta + C$

7. $= \frac{1}{3}x^{\frac{3}{2}}logx - \frac{2}{9}x^{\frac{3}{2}} + C$

8. $= \frac{3}{16}\pi$

# Integration Problems

1. $\displaystyle\int_1^{\sqrt{3}} \frac{dx}{(1+x^2)\sqrt{1+x^2}}$

2. $\displaystyle\int_0^1 \sqrt{2x-x^2}dx$

3. $\displaystyle\int_0^2 xe^{-x}dx$

4. $\displaystyle\int \left(\frac{e^x+e^{-x}}{2}\right)^2 dx$

5. $\displaystyle\int x^2e^{-x}dx$

6. $\displaystyle\int_0^{\frac{\pi}{3}} \frac{dx}{\cos^2 x}$

7. $\displaystyle\int x\sin x dx$

8. $\displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{cos\theta}{\sqrt{1+2sin\theta}}d\theta$

# Answers

1. $= \dfrac{\sqrt{3}-\sqrt{2}}{2}$

2. $= \dfrac{\pi}{4}$

3. $= 1 - \dfrac{3}{e^2}$

4. $= \dfrac{1}{4}(\dfrac{1}{2}e^{2x} + 2x - \dfrac{1}{2}e^{-2x}) + C$

5. $= -(x^2 + 2x + 2)e^{-x} + C$

6. $= \sqrt{3}$

7. $= -xcosx + sinx + C$

8. $= \sqrt{3} - \sqrt{2}$

# Integration Problems

1. $\displaystyle\int_0^1 \frac{x^2}{\sqrt{4-x^2}}dx$

2. $\displaystyle\int_0^2 \frac{x^2}{x^3+1}dx$

3. $\displaystyle\int \frac{1}{x\log x}dx$

4. $\displaystyle\int \frac{1}{1+e^x}dx$

5. $\displaystyle\int (x^2+2x)e^{2x}dx$

6. $\displaystyle\int \frac{1}{\cos^2\frac{x}{2}}$

7. $\displaystyle\int_0^{\frac{\pi}{2}} \sin^2 x dx$

8. $\displaystyle\int_0^{\pi} \theta sin\theta(cos\theta+\theta sin\theta)d\theta$

# Answers

1. $= \frac{\pi}{3} - \frac{\sqrt{3}}{2}$

2. $= \frac{2}{3} log3$

3. $= log|logx| + C$

4. $= x - log(e^x + 1) + C$

5. $= \frac{1}{4}(11e^4 - 3e^2)$

6. $= 2tan\frac{x}{2} + C$

7. $= \frac{\pi}{4}$

8. $= \frac{\pi^3}{6} - \frac{\pi}{2}$

# Integration Problems

1. $\int \frac{dx}{2\sqrt{x}}$

2. $\int \frac{x^3 - 4x^2 - x - 2}{x^2 - 5x + 4} dx$

3. $\int x^2 \log x dx$

4. $\int \frac{(\log x)^3}{x} dx$

5. $\int \log x dx$

6. $\int_0^{\pi} (sin\theta + cos\theta)^2 d\theta$

7. $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{d\theta}{\sin^2 \theta}$

8. $\int \frac{1}{\tan x} dx$

# Answers

1. $= \sqrt{x} + C$

2. $= \frac{(x+1)^2}{2} + 2log|\frac{x-1}{x-4}| + C$

3. $= \frac{x^3}{3}logx - \frac{x^3}{9} + C$

4. $= \frac{(logx)^4}{4} + C$

5. $= xlogx - x + C$

6. $= \pi$

7. $= \frac{2}{\sqrt{3}}$

8. $= log|sinx|C$

# Integration Problems

1. $\int \frac{x^3}{\sqrt{x^2+1}} dx$

2. $\int_1^{\sqrt{3}} \frac{dx}{(1+x^2)\sqrt{1+x^2}}$

3. $\int_1^e x \log(ex) dx$

4. $\int_0^2 e^x \sqrt{e^x} dx$

5. $\int x^2 e^{-x} dx$

6. $\int_0^{\frac{\pi}{2}} \frac{sin\theta}{1+cos\theta} d\theta$

7. $\int e^{2x} sin3x dx$

8. $\int sin^3\theta d\theta$

# Answers

1. $= \frac{1}{3}(x^2+1)^{\frac{3}{2}} - \sqrt{x^2+1} + C$

2. $= \frac{\sqrt{3}-\sqrt{2}}{2}$

3. $= \frac{3}{4}e^2 - \frac{1}{4}$

4. $= \frac{2}{3}(e^3-1)$

5. $= -(x^2+2x+2)e^{-x} + C$

6. $= log2$

7. $= \frac{1}{13}e^{2x}(2sin3x - 3cos3x) + C$

8. $= \frac{cos^3\theta}{3} - cos\theta + C$

# Integration Problems

1. $\int \sqrt{x^2+1}dx$

2. $\int_3^5 x(x-3)^2 dx$

3. $\int_1^e x^3 \log x dx$

4. $\int (x^2+2x)e^{2x} dx$

5. $\int (\log x)^2 dx$

6. $\int \sin(\pi x) dx$

7. $\int cos^3 x dx$

8. $\int_0^{\frac{\pi}{4}} \frac{(1+tanx)^3}{cos^2 x} dx$

# Answers

1. $= \frac{1}{2}(x\sqrt{x^2+1} + log(x + \sqrt{x^2+1})) + C$

2. $= 12$

3. $= \frac{3e^4+1}{16}$

4. $= \frac{1}{4}(11e^4 - 3e^2)$

5. $= x(logx)^2 - 2xlogx + 2x + C$

6. $= -\frac{1}{\pi}cos\pi x + C$

7. $= sinx - \frac{sin^3x}{3} + C$

8. $= \frac{15}{4}$

# Integration Problems

1. $\int_0^{\frac{\sqrt{2}}{4}} \frac{1}{\sqrt{1-2x^2}} dx$

2. $\int_1^2 \frac{x^2}{\sqrt{2x-1}} dx$

3. $\int \left(\frac{e^x + e^{-x}}{2}\right)^2 dx$

4. $\int x^2 e^{-x} dx$

5. $\int \frac{1}{1+e^{-x}} dx$

6. $\int sin^3 3\theta cos 2\theta d\theta$

7. $\int \frac{1}{\sin x} dx$

8. $\int_0^{\frac{\pi}{2}} (3cos\theta - cos3\theta)(cos\theta - cos3\theta) d\theta$

# Answers

1. $= \dfrac{\pi}{6\sqrt{2}}$

2. $= \dfrac{6}{5}\sqrt{3} - \dfrac{7}{15}$

3. $= \dfrac{1}{4}(\dfrac{1}{2}e^{2x} + 2x - \dfrac{1}{2}e^{-2x}) + C$

4. $= -(x^2 + 2x + 2)e^{-x} + C$

5. $= log(e^x + 1) + C$

6. $= \dfrac{1}{4}sin2\theta - \dfrac{1}{32}sin8\theta - \dfrac{1}{16}sin4\theta + C$

7. $= \dfrac{1}{2}log|\dfrac{1 - cosx}{1 + cosx}| + C$

8. $= \pi$

# Integration Problems

1. $\int \frac{x^3+1}{x^2-4}dx$

2. $\int_0^1 x\sqrt{1-x^2}dx$

3. $\int x^3 e^{x^2} dx$

4. $\int_{-1}^1 (x^2+1)e^{-x}dx$

5. $\int \log(x^2-4)dx$

6. $\int_0^\pi \sin 5x \sin 2x dx$

7. $\int \frac{sinx}{cos^2x+4sin^2x}dx$

8. $\int_0^{\frac{\pi}{4}} \frac{dx}{sin^2x+3cos^2x}$

# Answers

1. $= \frac{x^2}{2} + \frac{7}{4}log|x+2| + \frac{9}{4}log|x-2| + C$

2. $= \frac{1}{3}$

3. $= \frac{1}{2}(x^2-1)e^{x^2} + C$

4. $= 2e - \frac{6}{e}$

5. $= xlog(x^2-4) - 2x - 2log|\frac{x-2}{x+2}| + C$

6. $= 0$

7. $= \frac{1}{4\sqrt{3}}log\frac{2-\sqrt{3}cosx}{2+\sqrt{3}cosx} + C$

8. $= \frac{\sqrt{3}}{18}\pi$

# Integration Problems

1. $\int (x^2 + 2x + 3)(x + 1)dx$

2. $\int \frac{x^3 + 1}{x^2 - 4}dx$

3. $\int_1^e x \log(ex)dx$

4. $\int_0^1 \log(x + 1)dx$

5. $\int \frac{e^x}{(e^x + 1)^2}dx$

6. $\int \frac{sinx}{cos^2x + 4sin^2x}dx$

7. $\int x^2 \sin \pi x dx$

8. $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos^2 x dx$

# Answers

1. $= \frac{1}{4}(x^2 + 2x + 3)^2 + C$

2. $= \frac{x^2}{2} + \frac{7}{4}log|x + 2| + \frac{9}{4}log|x - 2| + C$

3. $= \frac{3}{4}e^2 - \frac{1}{4}$

4. $= 2log2 - 1$

5. $= -\frac{1}{e^x + 1} + C$

6. $= \frac{1}{4\sqrt{3}}log\frac{2 - \sqrt{3}cosx}{2 + \sqrt{3}cosx} + C$

7. $= (\frac{2}{\pi^3} - \frac{x^2}{\pi})cos\pi x + \frac{2}{\pi^2}xsin\pi x + C$

8. $= \frac{\pi}{6} - \frac{\sqrt{3}}{8}$

# Integration Problems

1. $\int (x^2 + 2x + 3)(x + 1)dx$

2. $\int_0^{\frac{1}{\sqrt{2}}} \frac{dx}{\sqrt{1 - x^2}}$

3. $\int \left(\frac{e^x + e^{-x}}{2}\right)^2 dx$

4. $\int (x - 1)e^{-x}dx$

5. $\int \frac{1}{1 + e^x}dx$

6. $\int_0^{\frac{\pi}{2}} \frac{cosx}{\sqrt{1 + sinx}}dx$

7. $\int xsin^2xdx$

8. $\int \frac{sinx}{cos^2x + 4sin^2x}dx$

# Answers

1. $= \frac{1}{4}(x^2 + 2x + 3)^2 + C$

2. $= \frac{\pi}{4}$

3. $= \frac{1}{4}(\frac{1}{2}e^{2x} + 2x - \frac{1}{2}e^{-2x}) + C$

4. $= -xe^{-x} + C$

5. $= x - log(e^x + 1) + C$

6. $= 2(\sqrt{2} - 1)$

7. $= \frac{x^2}{4} - \frac{x}{4}sin2x - \frac{1}{8}cos2x + C$

8. $= \frac{1}{4\sqrt{3}}log\frac{2 - \sqrt{3}cosx}{2 + \sqrt{3}cosx} + C$

# Integration Problems

1. $\displaystyle\int_1^{\sqrt{3}} \frac{dx}{(1+x^2)\sqrt{1+x^2}}$

2. $\displaystyle\int_1^{2} \frac{x^2}{\sqrt{2x-1}}dx$

3. $\displaystyle\int_e^{e^2} \frac{\log x}{x}dx$

4. $\displaystyle\int x(\log x)^2 dx$

5. $\displaystyle\int 2^x dx$

6. $\displaystyle\int_0^{\pi} \sqrt{1+cosx}dx$

7. $\displaystyle\int \tan^2 x dx$

8. $\displaystyle\int \tan x dx$

# Answers

1. $= \dfrac{\sqrt{3}-\sqrt{2}}{2}$

2. $= \dfrac{6}{5}\sqrt{3} - \dfrac{7}{15}$

3. $= \dfrac{3}{2}$

4. $= \dfrac{x^2}{2}\{(logx)^2 - logx + \dfrac{1}{2}\} + C$

5. $= \dfrac{2^x}{log2} + C$

6. $= 2\sqrt{2}$

7. $= tanx - x + C$

8. $= -log|cosx| + C$

# Integration Problems

1. $\displaystyle\int \frac{dx}{\sqrt{x}+\sqrt{x+2}}$

2. $\displaystyle\int_0^1 \sqrt{2x-x^2}dx$

3. $\displaystyle\int_1^e x\log(ex)dx$

4. $\displaystyle\int (x-1)e^{-x}dx$

5. $\displaystyle\int \log x dx$

6. $\displaystyle\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{sin^2 x}{cos^4 x}dx$

7. $\displaystyle\int \frac{1}{\sin x\cos x}dx$

8. $\displaystyle\int_0^{\frac{\pi}{2}} sin^3 x\cos^4 x dx$

# Answers

1. $= \frac{1}{3}((x+2)^{\frac{3}{2}} - x^{\frac{3}{2}}) + C$

2. $= \frac{\pi}{4}$

3. $= \frac{3}{4}e^2 - \frac{1}{4}$

4. $= -xe^{-x} + C$

5. $= xlogx - x + C$

6. $= \sqrt{3} - \frac{1}{3}$

7. $= log|tanx| + C$

8. $= \frac{2}{35}$

# Integration Problems

1. $\displaystyle\int \frac{dx}{2\sqrt{x}}$

2. $\displaystyle\int_0^1 \frac{x^5 + 2x^3}{(1+x^3)^3} dx$

3. $\displaystyle\int \sqrt{1+e^x} dx$

4. $\displaystyle\int \frac{(\log x)^3}{x} dx$

5. $\displaystyle\int \frac{1}{1+e^x} dx$

6. $\displaystyle\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x^2 - sinx) dx$

7. $\displaystyle\int \frac{1}{\sin x \cos x} dx$

8. $\displaystyle\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{sin^2 x}{cos^4 x} dx$

# Answers

1. $= \sqrt{x} + C$

2. $= \dfrac{7}{24}$

3. $= 2\sqrt{1+e^x} + log\dfrac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1} + C$

4. $= \dfrac{(logx)^4}{4} + C$

5. $= x - log(e^x + 1) + C$

6. $= \dfrac{\pi^3}{12}$

7. $= log|tanx| + C$

8. $= \sqrt{3} - \dfrac{1}{3}$

# Integration Problems

1. $\int x(x^2+1)^3 dx$

2. $\int (x^2+2x+3)(x+1)dx$

3. $\int x^3 e^x dx$

4. $\int_0^1 \sqrt{1+\left(\frac{e^x-e^{-x}}{2}\right)^2} dx$

5. $\int_0^2 xe^{-x} dx$

6. $\int \frac{sinx - cosx}{sinx + cosx} dx$

7. $\int (sinx + cosx)e^{-x} dx$

8. $\int e^{2x} sin3x dx$

# Answers

1. $= \dfrac{(x^2+1)^4}{8} + C$

2. $= \dfrac{1}{4}(x^2+2x+3)^2 + C$

3. $= (x^3 - 3x^2 + 6x - 6)e^x + C$

4. $= \dfrac{1}{2}(e - \dfrac{1}{e})$

5. $= 1 - \dfrac{3}{e^2}$

6. $= -log|sinx + cosx| + C$

7. $= -e^{-x}cosx + C$

8. $= \dfrac{1}{13}e^{2x}(2sin3x - 3cos3x) + C$

# Integration Problems

1. $\int \sqrt{x} dx$

2. $\int \frac{1}{x(x+2)} dx$

3. $\int_0^2 x e^{-x} dx$

4. $\int \log(x^2 - 4) dx$

5. $\int \frac{e^{3x} - 1}{e^x - 1} dx$

6. $\int_0^{\frac{\pi}{3}} \frac{dx}{\cos^2 x}$

7. $\int \sin(\pi x) dx$

8. $\int \sqrt{x} \log \sqrt{x} dx$

# Answers

1. $= \frac{2}{3}x^{\frac{3}{2}} + C$

2. $= \frac{1}{2}log|\frac{x}{x+2}| + C$

3. $= 1 - \frac{3}{e^2}$

4. $= xlog(x^2 - 4) - 2x - 2log|\frac{x-2}{x+2}| + C$

5. $= \frac{1}{2}e^{2x} + e^x + x + C$

6. $= \sqrt{3}$

7. $= -\frac{1}{\pi}cos\pi x + C$

8. $= \frac{1}{3}x^{\frac{3}{2}}logx - \frac{2}{9}x^{\frac{3}{2}} + C$

# Integration Problems

1. $\displaystyle\int \frac{\sqrt{x^2+1}}{x}$

2. $\displaystyle\int_0^1 \frac{x^2}{\sqrt{4-x^2}}dx$

3. $\displaystyle\int \frac{\log x}{x^3}dx$

4. $\displaystyle\int 2^x dx$

5. $\displaystyle\int \log x dx$

6. $\displaystyle\int \sin(\sqrt{x})dx$

7. $\displaystyle\int_0^\pi (sinx - \frac{1}{2}sin2x)^2 dx$

8. $\displaystyle\int \frac{1}{\cos x}dx$

# Answers

1. $= \sqrt{x^2+1} + log\frac{\sqrt{x^2+1}-1}{|x|} + C$

2. $= \frac{\pi}{3} - \frac{\sqrt{3}}{2}$

3. $= -\frac{2logx+1}{4x^2} + C$

4. $= \frac{2^x}{log2} + C$

5. $= xlogx - x + C$

6. $= -2\sqrt{x}cos\sqrt{x} + 2sin\sqrt{x} + C$

7. $= \frac{5}{8}\pi$

8. $= \frac{1}{2}log|\frac{1+sinx}{1-sinx}| + C$

# Integration Problems

1. $\int \sqrt{x} dx$

2. $\int_0^1 \frac{x^2}{\sqrt{4-x^2}} dx$

3. $\int \frac{\log x}{x^3} dx$

4. $\int x^3 e^{x^2} dx$

5. $\int (x^2 + 2x) e^{2x} dx$

6. $\int_0^{\frac{\pi}{2}} x^2 \cos x dx$

7. $\int \frac{1}{\cos^2 \frac{x}{2}}$

8. $\int (sinx + cosx) e^{-x} dx$

# Answers

1. $= \frac{2}{3}x^{\frac{3}{2}} + C$

2. $= \frac{\pi}{3} - \frac{\sqrt{3}}{2}$

3. $= -\frac{2logx + 1}{4x^2} + C$

4. $= \frac{1}{2}(x^2 - 1)e^{x^2} + C$

5. $= \frac{1}{4}(11e^4 - 3e^2)$

6. $= \frac{\pi^2}{4} - 2$

7. $= 2tan\frac{x}{2} + C$

8. $= -e^{-x}cosx + C$

# Integration Problems

1. $\displaystyle\int_1^{\sqrt{3}} \frac{dx}{(1+x^2)\sqrt{1+x^2}}$

2. $\displaystyle\int_0^{\frac{\sqrt{2}}{4}} \frac{1}{\sqrt{1-2x^2}}dx$

3. $\displaystyle\int_1^2 \log\frac{x+1}{2x}dx$

4. $\displaystyle\int \frac{1}{1+e^x}dx$

5. $\displaystyle\int \frac{\log x}{x^3}dx$

6. $\displaystyle\int_0^{\pi/3} \frac{dx}{1-\sin x}$

7. $\displaystyle\int_0^{\frac{\pi}{2}} \frac{cosx}{\sqrt{1+sinx}}dx$

8. $\displaystyle\int_0^{\frac{\pi}{4}} \frac{(1+tanx)^3}{cos^2x}dx$

# Answers

1. $= \dfrac{\sqrt{3}-\sqrt{2}}{2}$

2. $= \dfrac{\pi}{6\sqrt{2}}$

3. $= log\dfrac{27}{32}$

4. $= x - log(e^x + 1) + C$

5. $= -\dfrac{2logx + 1}{4x^2} + C$

6. $= \sqrt{3} + 1$

7. $= 2(\sqrt{2} - 1)$

8. $= \dfrac{15}{4}$

# Integration Problems

1. $\displaystyle\int_0^{\frac{1}{\sqrt{2}}} \frac{dx}{\sqrt{1-x^2}}$

2. $\displaystyle\int_0^1 x^2\sqrt{x}dx$

3. $\displaystyle\int_0^1 \log(x+1)dx$

4. $\displaystyle\int_1^2 \log\frac{x+1}{2x}dx$

5. $\displaystyle\int \frac{e^x - e^{-x}}{e^x + e^{-x}}dx$

6. $\displaystyle\int \frac{sinx}{cos^2x + 4sin^2x}dx$

7. $\displaystyle\int_0^{\frac{\pi}{2}} sin^3x\cos^4 xdx$

8. $\displaystyle\int x\sin xdx$

# Answers

1. $= \frac{\pi}{4}$

2. $= \frac{2}{7}$

3. $= 2log2 - 1$

4. $= log\frac{27}{32}$

5. $= log(e^x + e^{-x}) + C$

6. $= \frac{1}{4\sqrt{3}}log\frac{2-\sqrt{3}cosx}{2+\sqrt{3}cosx} + C$

7. $= \frac{2}{35}$

8. $= -xcosx + sinx + C$

# Integration Problems

1. $\int \sqrt[3]{(3x-1)^2}dx$

2. $\int_2^3 \frac{1}{1-x}dx$

3. $\int \frac{1}{1+e^x}dx$

4. $\int \frac{1}{e^x}dx$

5. $\int_0^1 xe^{-x^2}dx$

6. $\int_0^\pi \theta sin\theta(cos\theta + \theta sin\theta)d\theta$

7. $\int \frac{1}{\sin x \cos x}dx$

8. $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{sin^2 x}{cos^4 x}dx$

# Answers

1. $= \frac{1}{5}(3x-1)^{\frac{5}{3}} + C$

2. $= -log2$

3. $= x - log(e^x + 1) + C$

4. $= -e^{-x} + C$

5. $= \frac{1}{2}(1 - \frac{1}{e})$

6. $= \frac{\pi^3}{6} - \frac{\pi}{2}$

7. $= log|tanx| + C$

8. $= \sqrt{3} - \frac{1}{3}$

# Integration Problems

1. $\int \frac{x^3+1}{x^2-4}dx$

2. $\int \frac{dx}{(x-2)(x+1)^2}$

3. $\int_1^2 \log \frac{x+1}{2x}dx$

4. $\int \frac{1}{1+e^{-x}}dx$

5. $\int \log x dx$

6. $\int x cos^3 x dx$

7. $\int \tan^2 x dx$

8. $\int e^{2x} sin 3x dx$

# Answers

1. $= \frac{x^2}{2} + \frac{7}{4}log|x+2| + \frac{9}{4}log|x-2| + C$

2. $= \frac{1}{9}log|\frac{x-2}{x+1}| + \frac{1}{2(x+1)} + C$

3. $= log\frac{27}{32}$

4. $= log(e^x + 1) + C$

5. $= xlogx - x + C$

6. $= \frac{x}{4}(\frac{1}{3}sin3x + 3sinx) + \frac{1}{4}(\frac{1}{9}cos3x + 3cosx) + C$

7. $= tanx - x + C$

8. $= \frac{1}{13}e^{2x}(2sin3x - 3cos3x) + C$

# Integration Problems

1. $\displaystyle\int \frac{dx}{\sqrt{x}+\sqrt{x+2}}$

2. $\displaystyle\int_0^{\frac{1}{2}} \frac{x+1}{\sqrt{1-x^2}}dx$

3. $\displaystyle\int \frac{e^{2x}-1}{e^x-1}dx$

4. $\displaystyle\int \sqrt{1+e^x}dx$

5. $\displaystyle\int \frac{1}{e^x}dx$

6. $\displaystyle\int_0^{\frac{\pi}{2}} \frac{sin\theta}{1+cos\theta}d\theta$

7. $\displaystyle\int_0^{\frac{\pi}{3}} \frac{dx}{\cos^2 x}$

8. $\displaystyle\int_0^{\pi} (sin\theta+cos\theta)^2 d\theta$

# Answers

1. $= \frac{1}{3}((x+2)^{\frac{3}{2}} - x^{\frac{3}{2}}) + C$

2. $= -\frac{\sqrt{3}}{2} + 1 + \frac{\pi}{6}$

3. $= e^x + x + C$

4. $= 2\sqrt{1+e^x} + log\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1} + C$

5. $= -e^{-x} + C$

6. $= log2$

7. $= \sqrt{3}$

8. $= \pi$

# Integration Problems

1. $\displaystyle\int_1^2 \frac{dx}{(2x-1)\sqrt{2x-1}}$

2. $\displaystyle\int_0^1 (x^2+1)^3 dx$

3. $\displaystyle\int_0^{\sqrt{3}} \frac{x}{\sqrt{x^2+1}-x} dx$

4. $\displaystyle\int_1^e x^3 \log x dx$

5. $\displaystyle\int \frac{e^{3x}-1}{e^x-1} dx$

6. $\displaystyle\int sin^3\theta d\theta$

7. $\displaystyle\int_0^{\pi} \sin 5x \sin 2x dx$

8. $\displaystyle\int_0^{\frac{\pi}{2}} sin^3 x \cos^4 x dx$

# Answers

1. $= 1 - \frac{1}{\sqrt{3}}$

2. $= \frac{96}{35}$

3. $= \frac{7}{3} + \sqrt{3}$

4. $= \frac{3e^4 + 1}{16}$

5. $= \frac{1}{2}e^{2x} + e^x + x + C$

6. $= \frac{cos^3\theta}{3} - cos\theta + C$

7. $= 0$

8. $= \frac{2}{35}$

# Integration Problems

1. $\int_3^5 x(x-3)^2 dx$

2. $\int_1^2 \frac{dx}{1+\sqrt{x-1}}$

3. $\int x^3 e^x dx$

4. $\int \frac{e^{2x}-1}{e^x-1} dx$

5. $\int \frac{1}{1+e^{-x}} dx$

6. $\int \frac{1}{\sin x} dx$

7. $\int \sin(\pi x) dx$

8. $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{cos\theta}{\sqrt{1+2sin\theta}} d\theta$

# Answers

1. $= 12$

2. $= 2(1 - log2)$

3. $= (x^3 - 3x^2 + 6x - 6)e^x + C$

4. $= e^x + x + C$

5. $= log(e^x + 1) + C$

6. $= \frac{1}{2}log|\frac{1 - cosx}{1 + cosx}| + C$

7. $= -\frac{1}{\pi}cos\pi x + C$

8. $= \sqrt{3} - \sqrt{2}$

# Integration Problems

1. $\int_0^1 x^2\sqrt{x}dx$

2. $\int_2^3 \frac{1}{1-x}dx$

3. $\int_{-1}^1 (x^2+1)e^{-x}dx$

4. $\int \log(x^2-4)dx$

5. $\int x\log x dx$

6. $\int x\cos 2x dx$

7. $\int \frac{1}{\cos x}dx$

8. $\int_0^{\frac{\pi}{3}} \frac{dx}{\cos^2 x}$

# Answers

1. $= \frac{2}{7}$

2. $= -log2$

3. $= 2e - \frac{6}{e}$

4. $= xlog(x^2 - 4) - 2x - 2log|\frac{x-2}{x+2}| + C$

5. $= \frac{x^2}{2}logx - \frac{x^2}{4} + C$

6. $= \frac{x}{2}sin2x + \frac{1}{4}cos2x + C$

7. $= \frac{1}{2}log|\frac{1+sinx}{1-sinx}| + C$

8. $= \sqrt{3}$

# Integration Problems

1. $\int \frac{1}{\sqrt{x^2+1}} dx$

2. $\int x^2(x^3+1)^4 dx$

3. $\int \frac{(\log x)^3}{x} dx$

4. $\int \frac{\log x}{x^3} dx$

5. $\int \log x dx$

6. $\int cos^3 x dx$

7. $\int (cosx - xsinx) dx$

8. $\int_0^\pi (sin\theta + cos\theta)^2 d\theta$

# Answers

1. $= log(x + \sqrt{x^2 + 1}) + C$

2. $= \frac{1}{15}(x^3 + 1)^5 + C$

3. $= \frac{(logx)^4}{4} + C$

4. $= -\frac{2logx + 1}{4x^2} + C$

5. $= xlogx - x + C$

6. $= sinx - \frac{sin^3x}{3} + C$

7. $= xcosx + C$

8. $= \pi$

# Integration Problems

1. $\int_0^1 (x^2+1)^3 dx$

2. $\int x\sqrt{x+1}dx$

3. $\int 2^x dx$

4. $\int \frac{e^{3x}-1}{e^x-1}dx$

5. $\int \frac{2e^x+1}{e^x-e^{-x}}dx$

6. $\int_0^{3\pi/2} \cos(2x)dx$

7. $\int_0^{\pi/3} \frac{dx}{1-\sin x}$

8. $\int \frac{1}{\sin x \cos x}dx$

# Answers

1. $= \frac{96}{35}$

2. $= \frac{2}{5}(x+1)^{\frac{5}{2}} - \frac{2}{3}(x+1)^{\frac{3}{2}} + C$

3. $= \frac{2^x}{log2} + C$

4. $= \frac{1}{2}e^{2x} + e^x + x + C$

5. $= \frac{1}{2}log(e^x+1)|e^x-1|^3 + C$

6. $= 0$

7. $= \sqrt{3} + 1$

8. $= log|tanx| + C$

# Integration Problems

1. $\int \frac{x^2}{x-1} dx$

2. $\int_0^1 \frac{1-x}{\sqrt{x}+1} dx$

3. $\int_0^2 xe^{-x} dx$

4. $\int 2^x dx$

5. $\int \frac{e^{3x}-1}{e^x-1} dx$

6. $\int \frac{1}{\tan x} dx$

7. $\int_0^{\frac{\pi}{2}} \frac{cosx}{\sqrt{1+sinx}} dx$

8. $\int_0^{\frac{\pi}{2}} sin^3 x \cos^4 x dx$

# Answers

1. $= \frac{(x+1)^2}{2} + log|x-1| + C$

2. $= \frac{1}{3}$

3. $= 1 - \frac{3}{e^2}$

4. $= \frac{2^x}{log2} + C$

5. $= \frac{1}{2}e^{2x} + e^x + x + C$

6. $= log|sinx|C$

7. $= 2(\sqrt{2} - 1)$

8. $= \frac{2}{35}$

# Integration Problems

1. $\int \frac{x}{\sqrt{3-x^2}}dx$

2. $\int x(x^2+1)^3 dx$

3. $\int \frac{e^{2x}-1}{e^x-1}dx$

4. $\int_0^1 \sqrt{1+\left(\frac{e^x-e^{-x}}{2}\right)^2}dx$

5. $\int_0^1 xe^{-x^2}dx$

6. $\int sin^2 x \cos^2 x dx$

7. $\int \frac{1}{\sin x}dx$

8. $\int \sin(\pi x)dx$

# Answers

1. $= -\sqrt{3 - x^2} + C$

2. $= \dfrac{(x^2 + 1)^4}{8} + C$

3. $= e^x + x + C$

4. $= \dfrac{1}{2}(e - \dfrac{1}{e})$

5. $= \dfrac{1}{2}(1 - \dfrac{1}{e})$

6. $= \dfrac{1}{8}(x - \dfrac{1}{4}sin4x) + C$

7. $= \dfrac{1}{2}log|\dfrac{1 - cosx}{1 + cosx}| + C$

8. $= -\dfrac{1}{\pi}cos\pi x + C$